DS 574 **Stable Matching Worksheet**
Fall 2026 Prof. Kira Goldner

Stable Matching and School Choice

Definition 1. A *matching market* consists of a set A of n students and a set B of n schools, where each student a has a strict, complete preference order $\succ_a$ over B, and each school b has a strict, complete priority order $\succ_b$ over A. A *matching* M is a bijection between A and B; we write $M(a)$ for a's partner and $M(b)$ for b's partner.

Definition 2. A pair (a, b) with $M(a) \neq b$ is a *blocking pair* for the matching M if

that is, if a prefers b to its assigned partner and b prefers a to their assigned partner. A matching with no blocking pair is *stable*.

A blocking pair allows its two members to:

Deferred Acceptance

Other uses of DA. In the NRMP residency match, applicants propose to programs in rank order and programs hold tentative matches. New York City's 2003–04 high school redesign also used a version of DA with students proposing.

We start with students as proposers and schools as receivers. In the one-seat model we can also reverse the roles, with schools proposing in priority order and students holding their favorite offers. A proposal here is a step in the algorithm. The district sends the final offers.

Deferred Acceptance (DA), student-proposing

a. While some student a is unheld and has not yet proposed to every school:

(i) a proposes to the most-preferred school b on a's list among those not yet proposed to.

(ii) If b is holding nobody, then b *tentatively holds* a.

(iii) Otherwise b holds some a'; they keep whichever of a, a' they prefer and reject the other, who becomes unheld.

b. Output the matching M in which each school is assigned the student it holds.

Observations about DA (think: loop invariants). What patterns do you notice as the algorithm runs? Look at schools' held students, students' preference lists, and what must still be possible for an unheld student.

Theorem 1. *Deferred acceptance terminates after at most n^2 proposals, and the matching it outputs is a stable perfect matching. In particular, a stable matching always exists.*

Proof. Bound the number of proposals, then show that everyone is matched. For a possible blocking pair (a, b), consider separately whether a ever proposed to b. Which observation applies in each case?

Sanity check: how many proposals are made if every student has a different first choice?

Exercise. Run deferred acceptance with students proposing and then with schools proposing. Compare the partners.

$$a_1 : b_1 \succ b_2 \qquad b_1 : a_2 \succ a_1$$
$$a_2 : b_2 \succ b_1 \qquad b_2 : a_1 \succ a_2$$

The two outputs.

Student Optimality

Definition 3. School b is *achievable* for student a if there is some stable matching M with $M(a) = b$.

Theorem 2 (Gale–Shapley '62)**.** *Student-proposing deferred acceptance matches every student to their most-preferred achievable school. In particular, all students simultaneously get the best partner they could get in any stable matching, so there is a single* student-optimal *stable matching.*

Claim 1. Over the course of the algorithm, no student is ever rejected by an achievable school.

Why does the claim imply student optimality?

Proof of the claim. Suppose there is a first rejection by an achievable school: b rejects a in favor of a'. Fix a stable matching M with $M(a) = b$ and set $b' = M(a')$. Which schools have already rejected a'? Find a blocking pair for M.

Exercise. Where does the proof use that this is the first rejection by an achievable school?

School Choice

Students report preference lists. School priorities are fixed strict orders. A student–school pair blocks an assignment if the student prefers that school and it has a vacancy or an assigned student with lower priority. Schools may have several seats. With fixed strict priorities, a school tentatively holds its highest-priority applicants up to capacity and rejects the rest.

Theorem 3. *Student-proposing deferred acceptance is strategyproof for students: reporting a true preference list is a dominant strategy.*

Student optimality alone does not prove strategyproofness: a matching stable under a misreport need not be stable under the true preferences.

In the *Boston mechanism*, acceptances are final. In round k, each unassigned student applies to the school ranked kth; each school fills its remaining seats in priority order.
Immediate acceptance in practice. Boston used this rule before its 2005 decision to switch to DA. Chicago also used a version for selective high school admissions before its 2009 reform.

Exercise. Each school has one seat. Run immediate acceptance on these preferences. Can one student obtain a better school by changing their report?

Student	1st	2nd	3rd	Priority at	1st	2nd	3rd
s_1	Y	X	Z	X	s_1	s_3	s_2
s_2	X	Y	Z	Y	s_2	s_1	s_3
s_3	X	Y	Z	Z	s_3	s_2	s_1

Truthful outcome and a beneficial report (hold the other reports and all priorities fixed):

Is the truthful Boston outcome stable? What does student-proposing DA return?